

SURIYAN DHANAPAL

Software Engineer

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Summary

Software Engineer with **1.5+ years of hands-on experience** building **production-grade web and mobile applications**. Currently building three live products at Mindgryd – an **AI memory and context platform (Areev)**, an **agent-autonomy platform (Atmatic)**, and a **Pipedream-style automation platform (Axtion)** with **3,000+ connectors**. Strong in **full-stack development, backend system design, OAuth identity, and agentic automation**, with proven ability to own features end to end across backend and frontend and adapt to product pivots while keeping codebases clean and scalable.

Technical Skills

Programming Languages JavaScript, TypeScript, Python

Frameworks & Libraries Next.js, React, React Native (Expo), Node.js, Express, FastAPI

Backend & Databases RESTful API Design, Microservices Architecture, MongoDB, PostgreSQL, OAuth 2.0, Authentication and Authorization, Data Modeling

AI & Automation AI Agents, RAG / Retrieval, OCR, Workflow Automation, Prompt Engineering, Browser Automation

Tools & Platforms Git, GCP, Docker, GitHub Actions (CI/CD), Postman, VS Code, MongoDB Compass, NPM, PIP, Claude Code

Professional Experience

Software Engineer

October 2024 – June 2025

Hifresh Agro Private Limited

- Built an **internal delivery management mobile application from scratch** using Expo (React Native), fully owning the frontend architecture and UI while extending the existing Node.js / Express backend.
- Developed backend APIs for **delivery management, billing, and invoice generation** (Zoho Books API integration) and integrated **ESC/POS thermal printers** for instant in-app bill printing.
- Implemented **caching, stale-response handling, and structured logging** to cut redundant database queries, improve API response times, and strengthen reliability, with responsive layouts across phones and tablets.

Shifted to Mindgryd under the same founder, while HiFresh continues to operate; responsibilities unchanged

Software Engineer

June 2025 – Present

Mindgryd Software Private Limited

Areev (areev.ai) – AI memory and context platform for autonomous agents

- Designed and built the full **authentication and onboarding system** – OAuth 2.0 sign-in across **3 identity providers (Google, GitHub, Microsoft)** plus email/password, with organization auto-provisioning, JWT session validation, and role-based access control.
- Architected the **Knowledge Sources subsystem end to end (backend to frontend)** across **8 connectors** – Google Drive, Dropbox, Notion, OneDrive, SharePoint, Confluence, web search, and direct file upload – importing user content into a unified, retrievable knowledge layer (“grains”) with per-user OAuth credential isolation.
- Co-developed **CAL (Context Assembly Language)**, a declarative query language that fuses **vector, keyword, and graph retrieval in a single round-trip**, eliminating N+1 query fan-out across 10+ knowl-

edge primitives.

- Designed and built the **Hooks system** – a change-data-capture event layer delivering **HMAC-signed webhooks with exactly-once semantics** and exponential-backoff retries.

Atmatic (atmatic.ai) – *agent autonomy platform*

- Co-developing an **autonomy platform built on Areev** where AI agents **earn progressive autonomy through verified outcomes** – implementing **5 graduated autonomy levels and 8 decision-gate types** with full human-in-the-loop audit trails.
- Developed a **Next.js frontend** for configuration-heavy agent and workflow setup, balancing flexibility with usability.
- Built and integrated **AI-powered agents** for content generation, scheduling, caption rewriting, brand-kit extraction, prompt generation, and browser automation.
- Integrated **Razorpay and PayPal** payment gateways across secure frontend flows and backend transaction workflows, and designed **MongoDB schemas and core data models**.

Axtion – *internal workflow automation and integration platform*

- Co-developed an internal **Pipedream-style** automation and integration platform exposing **3,000+ first-party connectors**.
- Built and maintained a large share of the connector catalog, implementing provider-specific OAuth, API integrations, and data normalization across hundreds of third-party services.

Education

Bachelor of Engineering – Electronics and Communication Engineering

2019 – 2023

Dr. N.G.P. Institute of Technology

Projects

Cooper Compass – Agentic Document and Email Intelligence

IIT Bombay-Incubated Startup

coopercompass.com

Stack: Node.js, Express, TypeScript, OpenAI (gpt-4o-mini), Google Vision OCR, Tesseract.js, OAuth 2.0 (Gmail / Microsoft Graph), PostgreSQL, Redis / BullMQ, Docker, GitHub Actions

- Logistics teams spent 2+ hours manually extracting job and BL numbers from unstructured shipping emails – built an agentic email-intelligence platform under IIT Bombay incubation that **cut manual data entry from 2 hours to 8 minutes across 20+ pilot users**.
- Built a **Node.js / Express backend** for **real-time email ingestion** from Gmail and Outlook (OAuth 2.0 with Gmail Pub/Sub and Microsoft Graph push), with **Server-Sent Events** for live dashboard updates.
- Implemented the **document-extraction pipeline** Google Vision OCR with Tesseract.js fallback plus OpenAI classification backed by a regex layer extracting job/BL/AWB/invoice/container/port fields and auto-grouping every email and document by shipment at **2–5s end-to-end latency**.
- Deployed the full stack on client infrastructure with **Docker Compose and GitHub Actions CI/CD**, append-only migrations, and structured logging and monitoring.

Farm2Com – Expo React Native Mobile Application

Stack: React Native (Expo), Node.js, Express, MongoDB, REST APIs

- Developed an agriculture-focused mobile application featuring community posts, consultant chat, agricultural news, crop management, and user profiles.
- Integrated AI-based disease prediction and plant recommendation features to assist farmers with decision-making.